Telephone Modem Access Improves Diabetes Control in Those With Insulin-Requiring Diabetes

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OBJECTIVE — To assess whether modem access improves diabetes control in IDDM patients.

RESEARCH DESIGN AND METHODS — Forty-two patients participated in the study and were followed for 12 wk. The patients were randomly divided into two groups at baseline, a modem group and a control group. There were no significant differences between HbA_{1c}, random blood glucose, and weight between the groups at the beginning of the study. Patients were asked to perform five blood glucose determinations/day (before breakfast, before lunch, afternoon [1500], before dinner, and at bedtime) twice/week. The modem group transferred their data over the phone once/week. The control group would bring in their results on their regular visits every 6 wk. Patients in the modem group were counseled every week over the telephone after transferring results to adjust insulin and food intake if necessary.

RESULTS — In the modem group, HbA_{1c} improved from 0.106 to 0.092 (13.20%). The control group improved from 0.112 to 0.102 (8.9%). There was no significant change in weight, random blood glucose, or insulin.

CONCLUSIONS — The use of telephone modem–based patient-monitoring systems in diabetes clinical research seems to stimulate the patient to keep closer control of blood glucose levels. It might be especially useful in rural settings, for which this study was designed.

Self-monitored blood glucose is an essential part of diabetes self-management (1–11). However, several studies have indicated that, despite adequate knowledge, patients do not modify their insulin regimen or life-style determinants (1–4). Controlled trials do not appear to indicate that diabetes education significantly changes metabolic control unless there is positive reinforcement (1,2). In contrast, behavioral modification and setting of realistic goals appears to improve the outcome of diabetes education (4).

Assuming that easy access to diabetes care providers would allow patients to make appropriate adjustments to their management and thus improve their metabolic control, a controlled trial comparing customary management with continuous diabetes care provider access to patients self-monitored glucose data was undertaken. In this study, access was carried out through a telephone modem (Glucofacts Data-link telephone modem, Miles, Diagnostic Division, Elkart, IN) directly to stored data in the patients glucometer (Glucometer M, Miles). The telephone modem connection was therefore tested as a possible means of positive reinforcement. Newfoundland is a rural area of Canada with relatively long distances to the centers that provide diabetes educational programs. Thus, we inquired whether the use of telephone modems for the transmission of self-monitored blood glucose could present a distinct advantage to the management of the diabetic patient in rural communities.

RESEARCH DESIGN AND METHODS

Forty-two patients (22 women, 20 men; mean wt 77.8 kg [range 49.5–114.0 kg]) participated in the study. Patients were referred from two endocrine clinics within the Health Sciences Centre, St. John's, Newfoundland. The patients were consecutively chosen by one of the investigators, who did not know them.

Stratified blocked randomization was used to divide the patients into two groups at baseline to assure an even distribution between the study groups (13). The patients were randomly assigned to a control or modem (experimental) group; 22 patients were in the modem group and 20 in the control group. Four patients dropped out of the study, 2 from

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Table 1—*Questionnaire administered to modem group at the completion of study*

1. WAS THE PURPOSE OF THE STUDY EXPLAINED WELL ENOUGH TO YOU?
2. WAS THE USE OF THE MODEM EXPLAINED WELL ENOUGH TO YOU?
3. DO YOU FEEL THAT THE DIRECT CONTACT IMPROVED:
   A) BLOOD SUGAR CONTROL?
   B) KNOWLEDGE ABOUT DIABETES?
   C) KNOWLEDGE ABOUT INSULIN DOSAGE?
   D) MOTIVATION FOR BLOOD SUGAR CONTROL?
4. DID YOU GET ENOUGH COUNSELLING DURING THE STUDY?
5. WILL YOUR PARTICIPATING IN THE STUDY CHANGE YOUR ATTITUDE TO BLOOD MONITORING IN THE FUTURE?
6. IF YOU HAD THE CHOICE, WOULD YOU THEN PREFER THE MODEM CONTACT INSTEAD OF THE REGULAR CONTACT?

each group. One patient moved away, 1 died during cardiac bypass surgery, and 2 lost interest in continuing participation. Patients were invited to participate in the study if they met the following criteria: age 15–65 yr, HbA_{1c} ≥0.070, insulin-dependent diabetes, and ownership of a touch-tone telephone. Approximately 90% of the patients to whom the study was explained agreed to enter the study, knowing that they could end up in the control group. Six patients from the modem group lived in the St. John's area, within a 10- to 15-min drive from the hospital, 14 patients lived within a 15-min or 4-h drive from the hospital. The distribution of distance from the hospital was about the same in the control group. The follow-up period after entering the study was 12 wk.

Neither duration of disease nor age was significantly different ($P = 0.10$) between the modem and the control groups. Mean duration of diabetes was 11.93 ± 11.43 yr in the modem group and 11.19 ± 4.51 yr in the control group. The mean age of the modem group was 41.60 ± 16.93 yr (range 17–64 yr) and 41.25 ± 13.90 yr (range 20–65 yr) for the control group. There was no significant difference ($P = 0.10$) between HbA_{1c}, random blood glucose, or weight when comparing the modem and control groups at the beginning of the study.

Glucometer M (Miles, Elkart, IN) was used for the modem group. This glucometer gives a digital read out of capillary whole-blood glucose concentrations ranging from 1.4 to 22.1 mM (24 to 399 mg/dl). It can store up to 338 blood glucose test results with date and time. A telephone modem to fit Glucometer M is available. It allows the data stored in the glucometer to be transferred over telephone lines to a computer. An IBM XT compatible computer system featuring 640-kb RAM, 40-Mb hard disk, EGA screen, and parallel matrix printer was used to collect data. Glucofacts Data Management (Miles) software was used for all computer tasks. It includes the data-collecting program Data-Link and the data-processing program Glucofacts. In the control group, 18 patients used Glucometer M, 2 patients used Glucometer II (Miles), which is very similar to Glucometer M in terms of operation and range of readings.

All patients were asked to perform five blood glucose determinations/day (before breakfast, before lunch, afternoon [1500], before dinner, and at bedtime) twice/week. The modem group then transferred their data over the phone once a week. The data-link allows data to be collected in an assisted (manual) or unassisted (automatic) mode. All data collections in this study were done in an assisted mode because the telephone lines went through the hospital switchboard, which means that one of the investigators had to answer the phone. Only a direct line would have made the unassisted (automatic transfer without the telephone being answered) mode possible. The control group brought in their results either written down in diary form or saved in the memory of the Glucometer M if they happened to be using such a machine.

The data from the patients in the modem group were examined carefully by the investigators with Glucofacts, and the patients were counseled by telephone every week after transferring the results to adjust insulin dosage or food intake, if necessary. At the first visit, the patients in the experimental group were instructed on how to use the Glucofacts data-link telephone modem and glucometer M. They were all familiar with testing blood glucose accurately because they had been enrolled in the Diabetes Education Center of the hospital. For patients in the control group, no additional counseling to that customary was undertaken between the visits at 0, 6, and 12 wk.

When entering the study and at the end of each 6-wk period, all patients (control and experimental groups) attended clinic, had blood drawn to determine HbA_{1c} levels, and had their weight measured. After completing the study, the patients in the modem group were asked to fill out a questionnaire (Table 1). They were also asked to explain their answers to determine the background of their reply. Wilcoxon's rank-sum test (2 sided) was used for all statistical analyses because the data was nonnormally distributed (14). We considered a null hypothesis for all tests.

RESULTS — There was no significant difference ($P = 0.10$) between HbA_{1c}, random blood glucose, or weight between the groups at the beginning of the study. The experimental group made 203 transfers (mean ± SD 10.2 ± 1.81/patient). The total number of Glucometer M readings transferred was 2127 (mean 106.4 ± 43.65/patient). The total number of blood glucose results generated from the control group was 1301

Table 2—*HbA_{1c} when entering study*

	HbA_{1c}	
	0.070–0.110	>0.11
MODEM GROUP (*N*)	15	5
CONTROL GROUP (*N*)	8	10

χ^2 Test showed no significant difference between the distribution in the 2 groups.

(mean 72.27 ± 36.80/patient). The average number of measurements transferred each connection time was 10.5 ± 3.51. A few patients performed more tests than expected; these are also included in the average. The transmission time took on average ~2 min.

When the patients in the modem group came into the hospital for a prearranged visit, they did not transfer data that week. The data in their Glucometer M was then loaded directly into the computer, and the Glucometer M memory was cleared. Most transfers were successful, although a few subjects had problems with their telephone lines and modem signals. Some patients had to try twice before they succeeded in transferring data. Distance from the hospital did not have any bearing in the success of transfer. Indeed, a patient on a business trip successfully transferred data from Seattle, WA, to the study center.

Mean HbA_{1c} when entering the study was 0.106 ± 0.028 (range 0.082–0.208) for the modem group and 0.112 ± 0.018 (range 0.082–0.172) for the control group (Table 2). Normal range is 0.045–0.090. Because the distribution of HbA_{1c} was skewed, the results are most appropriately expressed as ranges rather than means ± SD. Fifteen patients had initial HbA_{1c} >0.110, and in 23 patients, HbA_{1c} was between 0.070 and 0.110 (Table 2). Over the 12-wk period, the number of hypoglycemic episodes was 112 (mean 5.6 ± 6.3/patient) in the modem group and 99 (mean 5.5 ± 3.4/patient) in the control group. The difference was nonsignificant. There were no episodes of hypoglycemic coma or diabetic ketoacidosis in either group.

In the modem group, the HbA_{1c} significantly improved after 6 wk (0.094 ± 0.012 vs. 0.106 ± 0.028, $P = 0.05$). After 12 wk in the study, the HbA_{1c} (0.092 ± 0.011) was further and significantly ($P = 0.05$) reduced compared with week 0. The control group showed a nonsignificant ($P = 0.10$) improvement in HbA_{1c} after both 6 and 12 wk compared with the start of the study (Table 3). The HbA_{1c} variations in the modem group from 1 to 6 wk were 11.32% and 13.20% from 1 to 12 wk. For the control group, the variations were 6.25% (1–6 wk) and 8.9% (1–12 wk). The change between the groups was significant after 6 wk but not after 12 wk.

There was a gradual but nonsignificant ($P = 0.10$) weight reduction in the modem group at 6 and after 12 wk compared with the start of the study (Table 3). In the control group, on the other hand, there was a gradual weight gain (Table 3). The random blood glucose levels in the modem group when entering the study were 8.59 ± 2.11 and 7.52 ± 2.15 mM after 6 wk and 7.67 ± 2.47 mM after 12 wk. For the control group, the initial random blood glucose levels were 8.86 ± 3.73 and 9.52 ± 2.96 mM after 6 wk and 7.97 ± 2.79 mM after 12 wk. The change in random blood glucose did not show any significant difference ($P = 0.10$) in either group. The mean insulin dose in the modem group at the beginning of the study was 43.25 ± 8.43 U/day and 45.50 ± 6.32 U/day in the control group. By the end of the study, insulin doses were increased to 44 U/day in the modem group and to 45.80 U/day in the control group. The insulin doses did not change significantly.

There were no correlations between education and motivation and between motivation and distance from the hospital. However, there seemed to be a correlation between age and motivation. Patients in the age-group between 34 and 54 yr obtained the best results. Eigh-

Table 3—*Values of HbA_{1c}, random blood glucose, and weight after 0, 6, and 12 wk*

	WEEK		
	0	6	12
HbA_{1c} (%)			
MODEM	0.106 ± 0.028 (0.070–0.208)	0.094 ± 0.012 (0.070–0.115)	0.092 ± 0.011 (0.080–0.110)
CONTROL	0.112 ± 0.018 (0.082–0.142)	0.105 ± 0.019 (0.080–0.140)	0.102 ± 0.012 (0.074–0.110)
RANDOM BLOOD GLUCOSE (MM)			
MODEM	8.6 ± 2.1	7.5 ± 2.2	7.7 ± 2.5
CONTROL	8.9 ± 3.7	9.5 ± 3.0	8.0 ± 2.8
WEIGHT (KG)			
MODEM	78.8 ± 16.3	78.5 ± 16.7	78.6 ± 16.7
CONTROL	76.5 ± 15.8	76.8 ± 15.6	77.0 ± 15.3

Values are means ± SD with ranges in parentheses.

Table 4—*Answers to questionnaire*

Question	Yes	No
1	18	0
2	18	0
3A	17	1
3B	12	6
3C	11	7
3D	16	2
4	16	2
5	12	6
6	12	6

teen of 20 questionnaires administered to the patients in the modem group were completed and returned. Based on the questionnaires, almost all of the patients suggested that their understanding of blood glucose control and motivation for self-management was improved. Approximately two-thirds of the patients felt their general knowledge about diabetes, e.g., long-term complications and insulin dosage, was improved. They also indicated that they were more aware of when to take the insulin. Many patients claimed that the study improved their understanding of the need for several measurements during the same day rather than single, random measurements. The patients who preferred regular contact with the physician rather than modem contact said that they were afraid of losing the personal contact with the doctor. A patient wrote that he tested more frequently because the glucostix were free.

CONCLUSIONS— This study demonstrates that direct diabetes care provider contact, i.e., with a telephone modem, had a beneficial influence on the control of diabetes compared with customary management. We found a significant difference in HbA_{1c} levels in the modem (experimental) group after 6 and 12 wk (compared with 0 wk) but failed to do the same in the control group. The drop in HbA_{1c} could not be explained on basis of weight reduction, decreased insulin doses, or random blood glucose. The lack of significant difference in blood glucose may reflect that patients in the control group may be more compliant on days that they monitor blood glucose. According to the questionnaires administered to the patients at the end of the study, the patients in the modem group gained new knowledge about and motivation for blood glucose control.

This study has the additional feature of being performed in a rural area of Canada with limited access to diabetes referral centers. Our results suggest that the telephone modem might be a practical way of providing specialist care in remote areas. However, the modem could also be beneficial in urban communities because it gives more reliable results than maintaining a diary. Regular analysis of the results and contact with a physician seems to motivate patients to put more effort into control of blood glucose.

The modem group experienced about twice as many hypoglycemic incidents compared with the control group, but the patients who had the highest number of hypoglycemic events were in general not the same patients who reduced HbA_{1c} significantly.

The cost of transmission was ~1.20 Canadian dollars every time, which, compared with the cost of transportation, consultation, and lost working time, is very inexpensive.

In general, the patients found it interesting and exciting to use this new equipment. They sometimes did extra measurements and asked additional questions about insulin and nutrition and made suggestions. The positive impact could be temporary, but we expect it to last because the response we received from patients was very positive and dynamic, which can explain the improvement in HbA_{1c}.

However, the data should encourage setting up a long-term study to see whether the benefit of telephone modems would persist. The fact that the glucostix were free during the study could also have a potential impact. If the high motivation continues, it will be a very practical and economical method to use for the health-care provider in diabetes care.

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References

1. Adamson ET, Gullion DS: Assessment of diabetes continuing medical education. *Diabetes Care* 9:11–16, 1986
2. Mazze RS, Pasmantier R, Murphy JA, Shamoon H: Self-monitoring of capillary blood glucose: changing the performance of individuals with diabetes. *Diabetes Care* 8:207–13, 1985
3. Mazzuca SA, Moorman NH, Wheeler ML, Norton JA, Fineberg NS, Vinicor F, Cohen SJ, Clark CM: The Diabetes Education Study: a controlled trial of the effects of diabetes patient education. *Diabetes Care* 9:1–10, 1986
4. Bloomgarden ZT, Karmally W, Metzger MJ, Brothers M, Nechemias C, Bookman J, Faierman D, Ginsberg-Fellner F, Rayfield E, Brown V: Randomized, controlled trial of diabetic patient education: improved knowledge without improved metabolic status. *Diabetes Care* 10:263–72, 1987
5. Saudek CD: Data source automation: new technology for the management of patient-generated test results. *Diabetes Med* 6:394–99, 1989
6. Burnett KF, Taylor CB, Agras WS: Computer-assisted management of weight, diet and exercise in the treatment of type II diabetes. *Diabetes Educ* 13:234–36, 1987
7. Langer O: Scientific rationale for management of diabetes in pregnancy: recent approaches with innovative computer-based technology. *Diabetes Care* 11:67–72, 1988
8. Albisser MA: Intelligent instrumentation in diabetic management. *Crit Rev Biomed Eng* 17:1–24, 1989
9. Williams CD, Harvey FE, Sonksen PH: A

role for computers in diabetes care? *Diabetic Med* 5:177–99, 1988
10. Zimmet P, Lang A, Mazze RS, Endersbee R: Computer-based patient monitoring systems: use in research and clinical practice. *Diabetes Care* 11:62–66, 1988
11. Rodbard D: Potential role of computers in clinical investigation and management of diabetes mellitus. *Diabetes Care* 11: 54–61, 1988
12. Holli BB, Calabrese RJ: *The Dietician's Guide*. Philadelphia, PA, Lea & Febiger, 1986
13. Hulley SB, Cummings SR: *Designing Clinical Research*. Baltimore, MD, Williams & Wilkins, 1988, p. 111–24
14. Kirkwood BR: *Essentials of Medical Statistics*. London, UK, Blackwell, 1988